



All-fiber microheater with attenuation-controlled hot-zone based on metal-doped silica fibers

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Abstract: We propose and experimentally demonstrate an optical microheater based on metal-doped optical fibers (MDFs) that exploits the intrinsic 1480 nm absorption of Co^{2+} and $\text{Fe}^{2+}/\text{Fe}^{3+}$ ions co-doped in an aluminosilicate core. Three MDFs with attenuation coefficients of 9.45, 12.58, and 32.43 dB/cm are fusion-spliced to single-mode fibers to form 10-mm-long heaters. Under 367 mW pumping, the peak surface temperatures reach 166, 268, and 478 °C, while the corresponding hot-zone lengths (FWHM) decrease from 10.6 to 2.8 mm, demonstrating that the attenuation coefficient is a single design parameter that simultaneously controls both temperature and spatial confinement. A Beer–Lambert distributed heat source coupled with a cylindrical fin model predicts the peak temperatures of the remaining samples within 5–21% from a single calibration point. Independent fiber Bragg grating measurements verify the thermographic results within 18 °C, and optically induced carbonization of an acrylate recoating confirms the practical heating capability. Unlike conventional coating-based fiber microheaters, the proposed device requires no external photothermal layer, is directly fusion-spliceable to standard SMFs, and enables attenuation-controlled tailoring of both the peak temperature and hot-zone length.

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1. Introduction

A microheater is a miniaturized heating device with dimensions of a few micrometers to several hundred micrometers, fabricated using conductive thin-film resistor technology, microelectromechanical systems (MEMS), or optical photothermal effects [1–4]. Owing to its ability to deliver fast and precise localized heating with low input power, microheaters have been widely employed in diverse fields, including responsivity enhancement of gas sensors, droplet ejection in inkjet printer heads, sample heating in polymerase chain reaction (PCR) instruments for biochemical analysis, and wearable thermal devices [5–8].

Optical microheaters generate localized heat at a target position by remote delivery of optical energy and are particularly suited to applications that are difficult to address with electrical microheaters, such as temperature control in microfluidic systems, thermal tuning of photonic devices, and precision thermal management in confined spaces [9–12]. Fiber-based architectures, in particular, allow physical separation between the light source and the heating element, thereby reducing wiring complexity, providing immunity to electromagnetic interference (EMI), and enabling easy insertion or embedding into narrow and harsh environments [13,14]. These features make fiber-optic microheaters a promising next-generation localized heat source platform that complements the structural limitations of conventional electrical microheaters [15].

Most fiber-optic microheaters reported to date rely on photothermal conversion layers—such as metal thin films, carbon nanotubes (CNTs), graphene, or silicon—deposited on the fiber tip [16–19]. However, these approaches require additional coating processes and are sensitive to variations in coating thickness, uniformity, and adhesion, which makes reproducibility difficult to ensure. Long-term operation also suffers from coating degradation and increased packaging complexity, limiting their extension to high-reliability and high-durability microheater applications.

In this paper, we propose and experimentally analyze an optical microheater that exploits the intrinsic optical absorption of metal-ion-doped silica fibers. Because the heating element is formed by a single fusion splice between a metal-doped fiber (MDF) and a standard single-mode fiber (SMF), the proposed structure fundamentally mitigates surface adhesion problems, simplifies the device geometry, and is compatible with mass production. Three MDFs with different optical attenuation characteristics are fabricated, and the peak surface temperature and the spatial extent of the heated region (hot zone) are systematically analyzed under 1480 nm photoexcitation using a high-performance thermal-imaging camera. The temperature distribution is theoretically predicted by combining a Beer–Lambert distributed heat source with a cylindrical fin heat-transfer model and is compared with experiment. The thermographic results are then independently validated using a fiber Bragg grating (FBG) attached to the microheater surface, and the practical heating capability is confirmed by optically induced thermal damage of an acrylate polymer recoating. Therefore, the present approach provides a coating-free, fully spliceable, and attenuation-engineered all-fiber microheater platform.

2. Theory

The optical power propagating along a metal-doped fiber attenuates exponentially in the propagation direction (z -axis) according to the Beer–Lambert law [20]:

$$P(z) = P_0 \exp(-\alpha z), \quad (1)$$

where P_0 is the input optical power and α is the absorption coefficient (Np/m). Experimentally measured values in dB/cm are converted to Np/m by the standard unit conversion [20]. The absorbed optical energy is assumed to be almost entirely converted into heat through non-radiative relaxation processes within the metal-doped fiber [21]. The heat-generation rate per unit length $q(z)$ [W/m] at position z is therefore given by [22]

$$q(z) = \alpha P(z) = \alpha P_0 \exp(-\alpha z). \quad (2)$$

Equation (2) describes an exponentially decaying distributed heat source that is maximum at the input end ($z = 0$) and minimum at the output end ($z = L$), and serves as the key input function of the present model.

The fiber, with a diameter $d = 125 \mu\text{m}$, is modeled as a thin cylindrical fin suspended in air [23]. Under steady-state conditions, the energy balance over an infinitesimal element dz consists of axial conduction, convective and radiative surface losses, and volumetric heat generation [23]:

$$-k A (d^2 T / dz^2) + h_{\text{eff}} \pi d (T - T_{\text{amb}}) = q(z), \quad (3)$$

where $T(z)$ denotes the local fiber surface temperature, T_{amb} is the ambient temperature, $k = 1.38 \text{ W/(m}\cdot\text{K)}$ is the thermal conductivity of silica [24], $A = \pi(d/2)^2$ is the cross-sectional area, and h_{eff} is the effective heat-transfer coefficient that accounts for both convection and radiation [25,26].

The characteristic decay length of axial conduction is given by [27]

$$\ell = \sqrt{k A / (h_{\text{eff}} \pi d)}. \quad (4)$$

Substituting the experimentally calibrated value of h_{eff} (Section 3.4) yields $\ell \approx 0.17 \text{ mm}$, which is more than two orders of magnitude smaller than the sample length $L = 10 \text{ mm}$. Therefore, except for a region of width $\pm\ell$ near the splice, the axial conduction term can be neglected and

Eq. (3) reduces to the local-equilibrium expression [22]

$$h_{eff} \pi d \Delta T(z) = q(z). \quad (5)$$

Combining Eqs. (2) and (5), the surface temperature distribution along the fiber takes the simple exponential form [22]

$$T(z) = T_{amb} + [\alpha P_0 \exp(-\alpha z)] / (h_{eff} \pi d). \quad (6)$$

Equation (6) implies that the optical absorption profile inside the fiber is directly mapped onto the surface temperature distribution. A larger absorption coefficient α thus concentrates the heat within a narrower region near the input end, giving the device a point-like heat-source character [28]. As described in Section 3.4, the only unknown parameter h_{eff} is determined from a single calibration measurement, and the calibrated value is used to predict the temperatures of the remaining samples.

3. Experimental results and discussion

3.1. Optical properties of the metal-doped fibers

Three types of MDFs (MDF01, MDF02, MDF03) were used in this study. The fibers were fabricated by combining the modified chemical vapor deposition (MCVD) process with a solution-doping technique. For solution doping, the porous silica soot layer deposited inside the substrate tube by MCVD was immersed in an ethanol-based precursor solution. Iron(III) chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$) and cobalt(II) chloride hexahydrate ($\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$) were used as the Fe and Co precursors, respectively, and were dissolved in ethanol for 2 h. The soot layer was soaked in the precursor solution for 1 h to impregnate the metal ions; the residual solution was then removed, and the tube was dried, sintered, and collapsed to form the doped-core preform. The core refractive index was tailored not by GeCl_4 but by AlCl_3 , which is added during solution doping to suppress aggregation of transition-metal ions [29]. Cobalt (Co) and iron (Fe) were incorporated into each core, and the optical absorption originates predominantly from Co^{2+} and $\text{Fe}^{2+}/\text{Fe}^{3+}$ ions. Co^{2+} in tetrahedral coordination produces a broad absorption band over 1300–1550 nm, whereas Fe^{3+} contributes mainly to losses in the visible region [30]. Hence, at the 1480 nm pump wavelength used in this work, heat generation is expected to arise predominantly from Co^{2+} absorption. Aluminum (Al) does not introduce any effective absorption band from the visible to the infrared and therefore does not directly contribute to optical absorption [31].

The elemental composition of each fiber core was analyzed by field-emission scanning electron microscopy (FE-SEM, SU8600, Hitachi Inc.) equipped with an energy-dispersive X-ray spectroscopy (EDS) detector. Cross-sectional samples were prepared by mounting the fibers vertically in epoxy, curing overnight, and polishing the end face; the core center was identified in the FE-SEM image, and EDS quantification was performed at that position. Table 1 lists the full EDS compositions, including O, Si, Al, Fe, and Co. The measured compositions confirm that MDF03 has the highest Co and Fe contents among the three fibers. The relative photothermal contributions of Co and Fe can be quantified from the molar absorptivities of the two ions in vitreous silica [30]. At 1480 nm, the molar absorptivity of Co is approximately $14 \text{ cm}^{-1}/(\text{mol} \cdot \text{L}^{-1})$, about 5.4 times that of Fe (approximately $2.6 \text{ cm}^{-1}/(\text{mol} \cdot \text{L}^{-1})$). Combining this ratio with the molar dopant ratios obtained from the EDS analysis (Table 1), Co accounts for essentially all of the absorption in the Fe-free MDF01 and for approximately 88% and 94% of the 1480 nm absorption in MDF02 and MDF03, respectively. Thus, although both ions absorb in the near-infrared, Co^{2+} is the dominant photothermal contributor at the pump wavelength in all three fibers.

The refractive-index profiles of the MDFs were measured using an IFA-100 multiwavelength optical fiber analyzer / fiber refractive-index profiler (Interfiber Analysis) at a measurement

Table 1. Elemental composition (wt%) of the Co/Fe-doped silica fiber cores obtained from EDS analysis.

Fiber	Co (wt%)	Fe (wt%)	Al (wt%)	O (wt%)	Si (wt%)
MDF01	0.17	0.00	3.82	50.09	45.92
MDF02	0.24	0.17	3.29	49.39	46.40
MDF03	0.83	0.28	3.10	49.60	46.19

wavelength of 633 nm. This measurement retrieves the transverse refractive-index distribution of the fiber from interferometric phase information rather than from longitudinal guided-power transmission at the 1480 nm pump wavelength. Therefore, the high optical attenuation of the MDFs is not expected to directly distort the measured refractive-index profiles. From the measured profiles, all three fibers were found to have identical core diameters of 8 μm . For calculating the optical parameters at the operating wavelength of 1480 nm, a silica cladding refractive index of 1.444 was used. The corresponding core refractive indices of MDF01, MDF02, and MDF03 were estimated to be 1.4525, 1.4512, and 1.4565, respectively (Table 2).

Table 2. Optical properties of the Co/Fe-doped silica fibers. Cladding refractive index $n_{\text{clad}} = 1.444$; core radius $a = 4 \mu\text{m}$; pump wavelength $\lambda = 1480 \text{ nm}$; V-number $V = (2\pi/\lambda) \cdot a \cdot \text{NA}$.

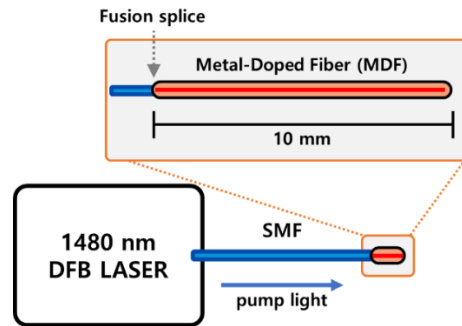
Fiber	Core index	V-number	α (dB/cm)
MDF01	1.4525	2.76	9.45
MDF02	1.4512	2.45	12.58
MDF03	1.4565	3.34	32.43

The V-numbers at the operating wavelength of 1480 nm were 2.76, 2.45, and 3.34 for MDF01, MDF02, and MDF03, respectively—values that exceed the single-mode cutoff of $V = 2.405$ and that, in principle, allow propagation of both the LP_{01} and LP_{11} modes. Because the MDFs were fusion-spliced without lateral offset to identical-core SMFs and were only 10 mm long, however, the LP_{01} mode is expected to be predominantly excited. The mode-field diameters (MFDs) calculated from the V-numbers are 8.2–8.7 μm —slightly smaller than that of a standard SMF ($\approx 10 \mu\text{m}$)—confirming that most of the optical field is confined within the core. The attenuation coefficients at 1480 nm were measured using an insertion-loss method in an SMF–MDF–SMF configuration. Each MDF was cleaved to a length of 1 cm and fusion-spliced between two standard SMFs. A 1480 nm DFB laser diode and an optical power meter were used to measure the transmitted power, and index-matching oil was applied to the bare MDF section to suppress cladding-guided light. The measured attenuation coefficients were 9.45, 12.58, and 32.43 dB/cm for MDF01, MDF02, and MDF03, respectively, with measurement errors of less than 0.5 dB/cm. The principal compositions and optical properties are summarized in Table 2.

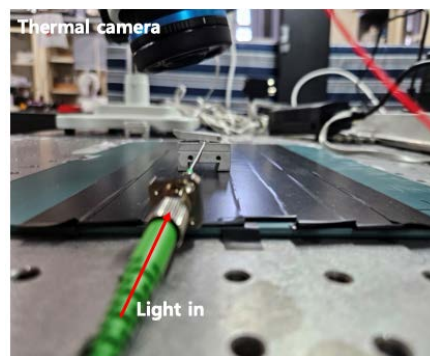
3.2. Microheater fabrication and measurement setup

Microheaters were fabricated by cleaving each MDF to a 10 mm length and fusion-splicing it to standard SMFs (Fig. 1). The same splicing parameters used for SMF–SMF splicing were applied without modification, ensuring fabrication simplicity.

Thermal characterization was performed using a 1480 nm distributed-feedback laser diode (QFBGLD-1480-500, QPhotonics Inc.) and an infrared thermal-imaging camera (FLIR A700) with $\pm 2^\circ\text{C}$ accuracy and 640×480 -pixel resolution. A close-up macro lens was attached to the camera so that the 125- μm -diameter, 10-mm-long fiber could be resolved with a spatial resolution of 2 μm per pixel. The emissivity was set to 0.95, which is an appropriate value for silica fibers,



(a)



(b)

Fig. 1. (a) Schematic of the metal-doped-fiber (MDF)-based optical microheater and (b) photograph of the thermographic measurement setup. The microheater is formed by fusion-splicing an MDF to an SMF; a 1480 nm DFB laser diode is used as the pump source, and the surface temperature distribution is measured with an infrared thermal-imaging camera.

and the working distance between the sample and the camera was 5.5 cm. The camera was tilted by approximately 30° to minimize spurious reflections from the bench surface. The measurement temperature range was 0–650 °C, and the maximum laser-diode output power was 367 mW. As shown in Fig. 1(b), the MDF was suspended 1.7 cm above the bench. All measurements were performed at 23 °C ambient temperature and 25% relative humidity, and were repeated five times per sample to assess reproducibility.

3.3. Thermographic measurement results

The thermal response of the MDF microheaters depended strongly on the optical attenuation. Figure 2 compares representative thermographic images and the axial temperature profiles obtained at the maximum input power (367 mW, 1480 nm). Distinct differences were observed not only in the peak temperature but also in the spatial distribution of the heated region. MDF01 produced a relatively long and broadly distributed heating zone, MDF02 displayed an intermediate behavior with stronger thermal localization, and MDF03 generated the highest peak temperature accompanied by a highly localized hotspot. This trend was consistent over the entire pump-power range and indicates that, even within the same 1 cm active section, increasing absorption

strengthens local optical absorption and concentrates the heat. Hence, the attenuation coefficient governs both the magnitude of heating and its spatial profile, and the operation of the MDF microheater can be switched between distributed heating and localized heating simply by tailoring the absorption.

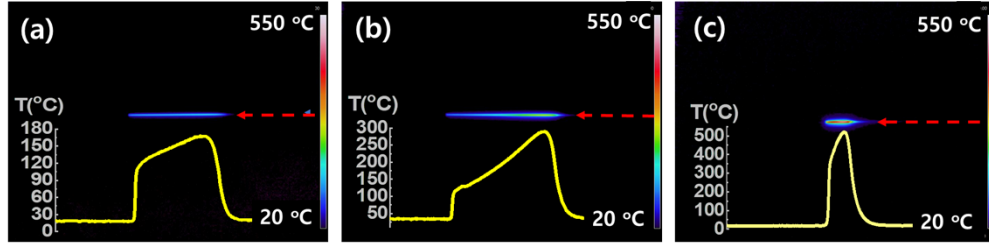


Fig. 2. Thermographic images (top) and axial temperature distributions (bottom) of the three MDF microheaters measured at the maximum input power (367 mW, 1480 nm): (a) MDF01 (9.45 dB/cm), (b) MDF02 (12.58 dB/cm), (c) MDF03 (32.43 dB/cm). The red dashed arrows indicate the direction of the input pump light.

Figure 3 shows the axial temperature distributions of the MDF microheaters as a function of input optical power. The temperature increased monotonically with the input power for all three samples. In particular, because MDF03 had the highest attenuation coefficient and absorbed the input light over a short distance near the input end, its effective heated region was confined to within approximately 5 mm—half the active length.

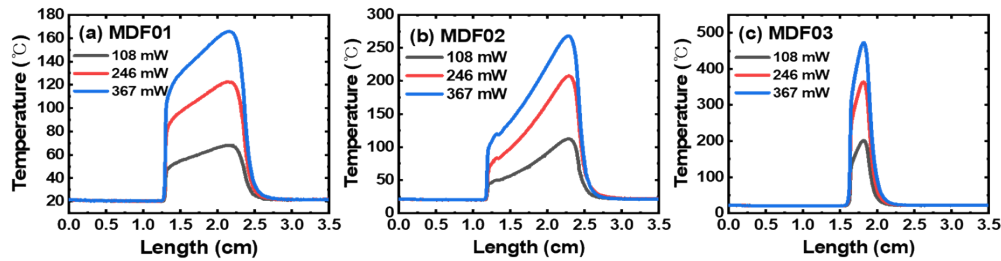


Fig. 3. Axial temperature distributions of the MDF microheaters at different input optical powers: (a) MDF01, (b) MDF02, (c) MDF03. Input optical powers: 108, 246, and 367 mW ($\lambda = 1480$ nm).

Figure 4(a) shows the maximum surface temperature and the corresponding standard deviation from five repeated measurements as a function of input optical power. For all three samples, the peak temperature increased with input power; the higher the attenuation coefficient, the more pronounced the saturation tendency at high powers. At the maximum input power of 367 mW, the average peak surface temperatures were approximately 166, 268, and 478 °C for MDF01, MDF02, and MDF03, respectively. The maximum standard deviation across the five repeated measurements was 4.3, 8.3, and 11.8 °C for MDF01, MDF02, and MDF03, respectively, exhibiting a slight increase in variability at higher attenuation and higher temperature. Nevertheless, these values do not significantly exceed the intrinsic accuracy of the thermal-imaging camera (± 2 °C), confirming the overall good thermal stability of the proposed microheaters.

The hot-zone length was defined as the full width at half maximum (FWHM) of the axial temperature profile in Fig. 3. Figure 4(b) shows the hot-zone length and its standard deviation as a function of input optical power. MDF01 maintained the longest hot zone (≈ 10.3 – 10.6 mm) over the entire power range, with the entire 10 mm active section acting as a heated region. MDF02

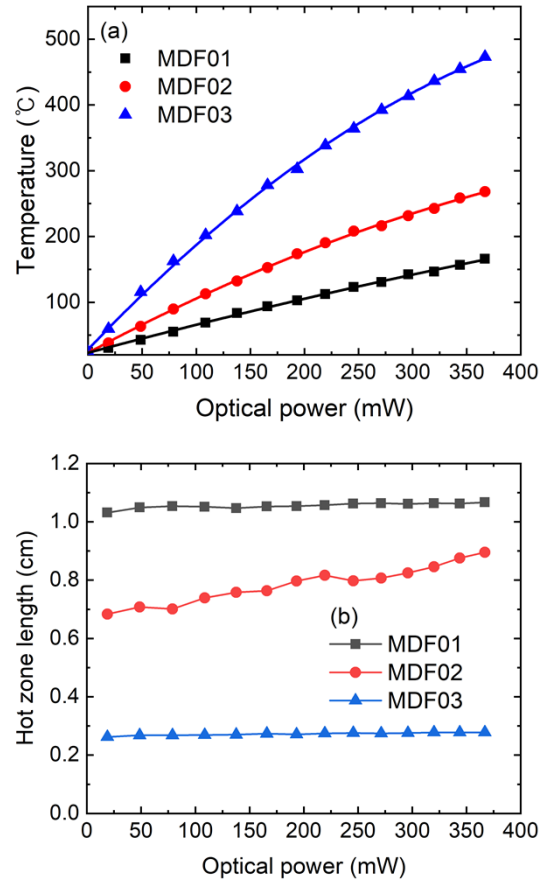


Fig. 4. (a) Average peak surface temperature and (b) hot-zone length (FWHM) of the MDF microheaters, and standard deviation from five repeated measurements as a function of input optical power. Second-order polynomial fits are also shown in (a).

exhibited a hot-zone length that increased from approximately 6.8 mm at low input power to approximately 8.9 mm at the maximum input power, indicating that the heated region of MDF02 can be tuned by adjusting the input power. MDF03 had the shortest and most localized hot zone (≈ 2.6 – 2.8 mm), which remained nearly constant regardless of pump power. These results demonstrate that highly absorbing fibers behave as concentrated, hotspot-like heat sources. The standard deviation of the hot-zone length was within 0.1 mm for MDF01 and MDF03; for MDF02, in contrast, it varied from 0.1 to 0.45 mm because the hot-zone length itself changed strongly with input power. Overall, an increase in the attenuation coefficient simultaneously raises the peak temperature and shortens the heated region, confirming that the attenuation coefficient is a primary design parameter that controls both the temperature and the spatial confinement of the heating zone of the MDF microheater.

Table 3 compares the proposed device with representative fiber-optic microheaters. Unlike previous tip- or coating-based heaters [16–19], which rely on an external photothermal coating or a discrete absorbing element at the fiber end-face, the present MDF microheater generates heat within the doped silica core itself. It therefore requires no external coating—removing coating thickness, uniformity, adhesion, and degradation as reliability concerns—and is directly fusion-spliced inline between standard SMFs. Moreover, a single design parameter, the attenuation

coefficient, simultaneously tunes the peak temperature (166–478 °C at 367 mW) and the hot-zone length (FWHM 2.8–10.6 mm), enabling continuous selection between distributed and localized heating within one inline geometry.

Table 3. Comparison of representative fiber-optic microheaters with the present work.

Ref.	Heater type / fabrication	Pump λ , power	Max. temp.	Heated region	Coating-free / spliceable
[16]	Absorber-coated pulled fiber tip; electrode-puller + tip coating	Red laser, 75 mW	~41 °C	Tip, 20–100 μm	No / No
[17]	CNT layer on capillary-fiber tip; laser-induced deposition	980 nm, >34 mW	Numerical only	Tip / end-face	No / partial
[18]	Graphene-acrylate coating (~20 μm); in-draw coating	Thermal sensing	~104 °C	Distributed	No / fiber
[19]	Crystalline-Si Fabry-Pérot element; fusion-spliced to endface	980 nm LD	>1000 °C	Tip, 100 $\mu\text{m} \times 10$ –200 μm	Discrete Si / yes
This work	Co/Fe-doped silica core (intrinsic); MCVD + solution doping, inline splice	1480 nm, 367 mW	478 °C (MDF03)	2.8–10.6 mm (FWHM), α-tunable	Yes / yes

3.4. Validation of the temperature-prediction model

The Beer–Lambert distributed heat source–cylindrical fin model proposed in Section 2 was applied to the experimental data. Using the measured peak surface temperature of MDF01 (9.45 dB/cm) at 367 mW ($T_{\text{max}} = 166$ °C, $T_{\text{amb}} = 23$ °C), the effective heat-transfer coefficient was calibrated via Eq. (6) to be $h_{\text{eff}} \approx 1,451$ W/(m²·K). The corresponding axial conduction decay length, calculated from Eq. (4), is $\ell \approx 0.17$ mm—more than two orders of magnitude smaller than $L = 10$ mm—confirming the validity of the local-equilibrium assumption.

Using the calibrated h_{eff} , the predicted maximum surface temperatures of MDF02 (12.58 dB/cm) and MDF03 (32.43 dB/cm) at the same input power are approximately 210 °C and 501 °C, respectively. Compared with the measured values of 268 °C and 478 °C, the prediction errors are approximately 21% [= (268–210)/268] for MDF02 and 5% [= (501–478)/478] for MDF03. The larger discrepancy for MDF02 than for MDF03 is mainly attributed to the simplified structure of the model and the intermediate heating regime of MDF02. In the model, the predicted peak temperature approximately linearly with the input-end attenuation [Eq. (6)], so any longitudinal non-uniformity in the dopant or absorption profile can strongly affect the effective near-input heat generation. In addition, MDF02 operates in a transitional regime: its hot-zone length increases with pump power [Fig. 4(b)], whereas MDF01 and MDF03 remain nearly constant. The single- h_{eff} local-equilibrium model is therefore most representative for the two limiting cases of distributed heating (MDF01) and strongly localized heating (MDF03), and less accurate for the intermediate MDF02 case. The model underprediction for MDF02 (210 vs. 268 °C) suggests that its actual near-input absorption may be higher than the length-averaged attenuation value. Even so, the calibrated h_{eff} obtained from a single reference point captures both the qualitative and the quantitative thermal behaviors of samples with different attenuation coefficients, demonstrating the predictive capability of the proposed model and the consistency of the thermographic measurements.

3.5. Validation of thermographic measurements using a surface-attached FBG

To independently verify the surface temperature measured by the thermal-imaging camera, an FBG—commonly used as an optical temperature sensor—was employed to measure the

temperature of the same sample. Cross-comparison between the two measurement techniques was used to assess the reliability of the obtained temperature values.

A 10-mm-long FBG, identical in length to MDF03, was placed in contact with the microheater and thermally coupled using a thin layer of boron-nitride paste (Slice Engineering Inc.; thermal conductivity 31.4 W/(m·K), temperature rating 850 °C). A high-temperature bonding agent was required because MDF03 reaches up to ≈ 480 °C, well above the thermal limit of common adhesives. Figure 5(a) shows the FBG reflection spectra of MDF03 as a function of input optical power. Because the microheater exhibits a highly non-uniform axial temperature distribution [Fig. 3(c)], the Bragg wavelength does not shift uniformly; instead, thermal chirping occurs and a multi-peak structure appears in the reflection spectrum because of localized Fabry–Pérot resonances formed at the boundaries between heated and unheated regions [32]. The longest-wavelength peak in this interference spectrum corresponds to the FBG section located in the highest-temperature region of the microheater, and tracking this wavelength yields the peak temperature.

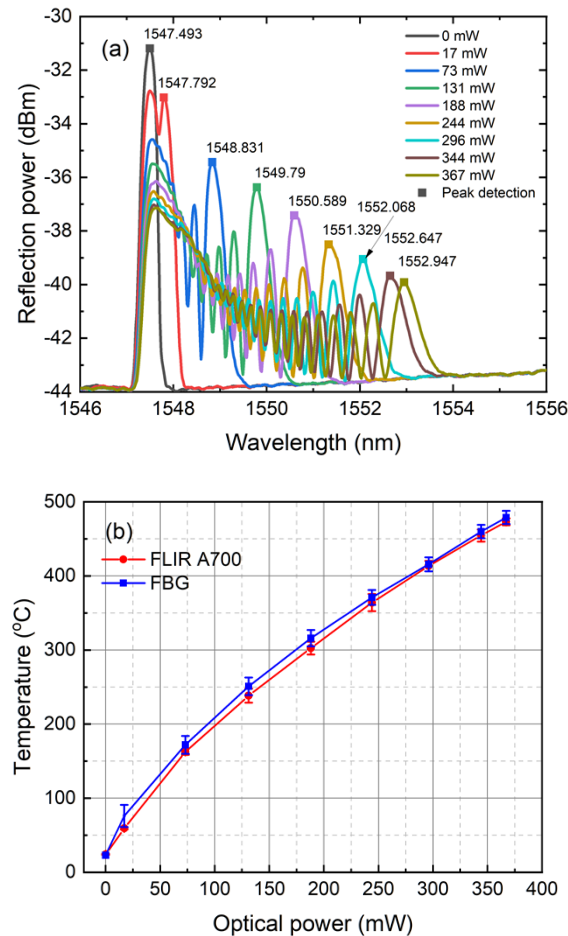


Fig. 5. Temperature measurements using an FBG attached to the MDF03 microheater. (a) FBG reflection spectra at different input optical powers. (b) Comparison between the apparent peak temperature deduced from the FBG wavelength shift and the maximum surface temperature measured by the thermal-imaging camera.

The microheater temperature was estimated from the spectral evolution of the FBG attached to the microheater surface. The initial Bragg wavelength measured at room temperature (23 °C) was defined as the reference wavelength λ_0 . When optical power was applied, the FBG spectrum did not shift uniformly as a single peak but instead broadened toward longer wavelengths, indicating non-uniform axial heating rather than uniform heating of the entire 10 mm grating. Therefore, the maximum local temperature of the microheater was estimated from the long-wavelength edge of the broadened spectrum. The maximum wavelength shift is defined as [33–35]

$$\Delta\lambda_{max} = \lambda_{edge} - \lambda_0 = \int_{T_0}^{T_{max}} S_T(T) dT, \quad (7)$$

where λ_{edge} denotes the edge wavelength of the long-wavelength reflection component. Because the temperature sensitivity of an FBG at high temperatures differs from the room-temperature value of 10 pm/°C, a temperature-dependent FBG sensitivity $S_T(T)$ was used rather than a single constant [33–35]. The temperature-segmented sensitivities adopted were 10 pm/°C for 23–100 °C, 11.8 pm/°C for 100–200 °C, 13.3 pm/°C for 200–300 °C, 14.4 pm/°C for 300–400 °C, and 15.1 pm/°C for 400–500 °C [35]. The temperature obtained in this way should be interpreted as the apparent peak temperature observed by the FBG, since the wavelength shift can include a small strain component arising from thermal expansion of the bonding layer and the heater structure, in addition to the pure temperature effect.

At 367 mW, the maximum FBG wavelength shift was approximately 5.9 nm, corresponding to an apparent peak temperature of approximately 460 °C when the temperature-dependent FBG sensitivity was taken into account. This value is close to the peak surface temperature of 478 °C measured by the thermal-imaging camera under the same conditions. Figure 5(b) compares the FBG-derived apparent peak temperature with the peak surface temperature obtained from the thermal-imaging camera as a function of input optical power. The two measurements show the same increasing trend and remain in reasonable agreement over the investigated power range, supporting the reliability of the thermographic measurement. However, the FBG-derived temperature should be interpreted as an apparent peak temperature rather than a purely strain-free temperature. The boron-nitride paste used here is a heat-transfer and anti-seize paste, not a rigid structural adhesive, so strain transfer from the heater to the FBG is expected to be limited. Nevertheless, a small strain contribution cannot be completely excluded. Therefore, the FBG result is used as an independent consistency check of the thermographic measurement rather than as an absolute temperature reference.

To verify the practical heating capability of the microheater, an MDF03 sample re-coated with a 250- μ m-thick acrylate polymer (LS2211, LuvantixADM Inc.) was exposed to the maximum input optical power for 2 minutes (Fig. 6). Clear physical thermal damage was observed in the coating layer over the heated zone ($\approx$ 2 mm hot zone). The image shows a concentric color change progressing from yellow to brown to black around the center of the heated region, visually demonstrating that the heat generated by optical absorption propagates outward from the heating center and induces stepwise thermal decomposition and carbonization of the coating material. The total length of the thermally affected region was approximately 3.7 mm, while the fully carbonized region of intense heat exposure was approximately 3 mm. Within the maximum available output power of 367 mW, no structural damage, splice failure, or performance degradation of the MDF microheater itself was observed; the device failure threshold therefore lies above 367 mW. The damage shown in Fig. 6 is confined to the external acrylate recoating and occurred after 2 min of exposure at 367 mW, where the bare-fiber peak surface temperature reaches approximately 478 °C. Thus, Fig. 6 demonstrates the local heating capability of the device, which is sufficient to carbonize an external polymer coating, rather than the failure limit of the MDF microheater itself.

That an external coating material can be physically damaged by purely optical means—without any electrical wiring—experimentally confirms that the proposed fiber-optic microheater can

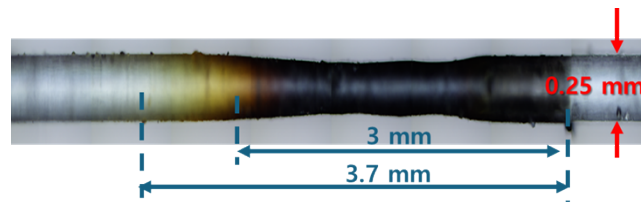


Fig. 6. Microscope image of thermal damage to a 250- μm -thick acrylate polymer recoating on the MDF (metal-doped fiber). The microheater was exposed to the maximum input optical power for 2 minutes. The observed damage corresponds to thermal carbonization of the external acrylate recoating, not structural failure of the MDF microheater.

serve as an effective heat source for applications that require precise local thermal processing, such as thermal ablation, localized material processing, and thermal curing. Combined with the inherent capability of optical fibers to be inserted into narrow spaces, the proposed device shows strong potential for selective localized heating in environments inaccessible to conventional electrical microheaters.

4. Conclusion

We have fabricated three 1-cm-long MDF microheaters with different optical attenuation characteristics and have systematically evaluated their thermal characteristics under 1480 nm photoexcitation. The measured attenuation coefficients were 9.45, 12.58, and 32.43 dB/cm for MDF01, MDF02, and MDF03, respectively. Thermographic measurements and axial temperature-profile analysis confirmed that the heating characteristics depend strongly on the attenuation coefficient.

As the attenuation increased, the peak surface temperature rose markedly while the hot-zone length decreased. At the maximum input power of 367 mW, the average peak surface temperatures reached approximately 166, 268, and 478 °C for MDF01, MDF02, and MDF03, respectively, with corresponding hot-zone lengths of approximately 10.6, 8.9, and 2.8 mm. These results demonstrate that the attenuation coefficient of the MDF is an effective design parameter that allows distributed and localized heating regimes to be addressed independently within an essentially identical device geometry.

The Beer–Lambert distributed heat source–cylindrical fin model predicted the peak temperatures of the remaining samples from a single calibration point with errors of approximately 5% for MDF03 and 21% for MDF02. To verify the reliability of the thermographic measurements, an independent FBG-based temperature measurement was carried out on MDF03; at the maximum input power of 367 mW, the FBG-deduced peak temperature was approximately 460 °C, differing by only ≈ 18 °C from the 478 °C measured by the thermal-imaging camera. In addition, optically induced thermal decomposition and carbonization of an acrylate polymer recoating directly demonstrated the practical heating capability of the device.

In summary, this work demonstrates that the heating temperature and the spatial confinement of the hot zone of an optical microheater can be controlled simultaneously by tailoring the attenuation of a metal-doped optical fiber that is fusion-spliced to standard SMFs. The proposed structure benefits from a simple fabrication process, excellent thermal stability and reliability, and seamless integration with conventional fiber-optic systems. Future work will focus on optimizing the metal-doping concentration, tailoring the active length, selecting the excitation wavelength, and implementing closed-loop temperature control to extend the device to higher operating temperatures and a broader range of thermal functions. We believe that the proposed MDF microheater offers a versatile and scalable platform for thermal ablation, localized polymer

curing, and precision heating in confined or electromagnetically harsh environments where conventional electrical microheaters are difficult to deploy. Compared with previously reported fiber-optic microheaters based on externally deposited photothermal coatings or discrete absorbing elements, the present MDF microheater provides a simpler and more robust all-fiber geometry because the heat is generated inside the doped silica core itself. This coating-free and fully spliceable configuration reduces concerns related to coating adhesion, thickness nonuniformity, and degradation while preserving compatibility with conventional fiber-optic systems.

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